

SARA SHARMA

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Professional Summary

AI/ML Engineer with experience building machine learning solutions across research, policy, and industry. Skilled in developing predictive models, working with large and varied datasets, and evaluating model performance. Experience includes research at IIT Bombay, analytics at NITI Aayog, and AI/ML research under DRDO guidance.

Work Experience

- Indian Institute of Technology Bombay** | *AI/ML Intern* Sept 2026 – Present
- Building an end-to-end machine learning research project involving model training, dataset curation, and spatial scene understanding for architectural applications.
 - Working across dataset preparation, model development, and evaluation for spatially structured data.
- NITI Aayog** | *Data Management & Analytics Intern* Aug 2026 – Present
- Developed prescriptive machine-learning and forecasting solutions for analysis of datasets supporting policy decisions.
 - Translated policy objectives into data-driven analytical solutions through requirements analysis and process evaluation.
- Fluentgrid Ltd.** | *Python & Machine Learning Intern* June 2025 – Aug 2025
- Engineered a multi-engine XGBoost system incorporating forecasting, risk analysis and scenario analysis for supply-chain intelligence.
 - Implemented SHAP-based model interpretability to examine model behaviour and support transparent, data-driven decisions.
- Inferrix Greentech** | *AI Research Intern* Jul 2024 – Sep 2024
- Developed a Physics-Informed Neural Network (PINN) integrating HVAC, weather, building and real-time IoT data for predictive energy modelling.
 - Incorporated heat-load profiles into the deep-learning architecture, achieving a reported 15% reduction in building energy consumption.

Projects

- When Should an ML System Not Trust Itself?** | *Evaluating ML Reliability Under Distribution Shift [Python · XGBoost]*
- Developed a framework to evaluate ML reliability under temporal and spatial distribution shift using uncertainty, calibration & OOD detection.
 - Compared standard, uncertainty-aware and abstention-based models to analyze failure modes and risk-performance trade-offs.
 - Extended evaluation to LLMs under adversarial and out-of-distribution inputs.
- AI-Driven Supply Chain Intelligence Platform** [Python · Scikit-learn · XGBoost · SHAP · Next.js]
- Engineered a 5-engine XGBoost system (forecasting, risk, scenario analysis) to optimize supply chain visibility.
 - Implemented SHAP analysis to ensure model interpretability, enabling data-driven stakeholder decision-making.

Volunteer | Defence Research & Development Organization

- AI/ML Researcher** | DRDO — Scientist-Supervised Research 2026 – Present
- Developing an AI/ML model for defence applications under the technical guidance and approval of a DRDO scientist.
 - Researching AI workflows integrating heterogeneous data modalities for defence-oriented applications.

Research & Publication

- "ECOAVIA"- AI-Driven Hybrid Forecasting for Sustainable Aviation & Development** | ICISC 2026 [\[Link\]](#)
- Designed an ensemble model (XGBoost, ARIMA, LSTMs) integrating climate and infrastructure data to forecast aviation demand and support sustainable policy advisory.

Education

- SRM Institute of Science and Technology** Chennai, TN
- Bachelor of Technology (B.Tech) in Computer Science and Engineering (AI & ML) | CGPA: 8.18/10* 2026

Technical Skills

- Programming:** Python, SQL, JavaScript
- Machine Learning:** PyTorch, TensorFlow, Scikit-learn, XGBoost, Neural Networks, Deep Learning
- Research & Modeling:** Statistical Modeling, Model Evaluation, Anomaly Detection, Explainable AI, SHAP, Time-Series Modeling
- AI:** Multimodal AI, NLP, RAG, LLM Applications
- Data:** NumPy, Pandas, PostgreSQL, ETL
- Engineering:** FastAPI, Docker, Git, CI/CD, MLflow